Records

MSSE 610 Software Requirements and Processes

From Week 2, every project assignment is two files submitted together: that week's document, and this one.

Team	NCM Innovators (Minnette Johnson, Carolann Mora, Peter Olayiwola)
Week and assignment	Week 3 – Use-Case Model and Conflicts / Plan-Driven Analysis
Package version	v1.2

Decision Record

Conflict 1 — Team approved; product owner approval pending for Week 4.

The question	How should the system support future rental booking (FR-03/FR-04) without misrepresenting readiness when a bike is overdue, uninspected, in workshop or otherwise blocked (FR-18/FR-19/FR-07)?
Positions	Proposal A: confirm on expected returns to preserve sales. Proposal B: delay confirmation until actual inspection to avoid false promises.
Resolution	Team approval: On November 7, 2026, Minnette Johnson and Carolann Mora both approved Peter Olayiwola's automated waitlist and optional free upgrade proposal in the team's Slack discussion about Conflict 1, as reported by Peter. Product owner approval: pending for Week 4. Decision: When a projected return is the only evidence, acknowledge the customer's waitlist request by email or SMS without confirming a booking, promising an inspection time, or taking a deposit. Confirm only after a suitable bike is verified for the requested period and normal booking and deposit steps succeed. If the original bike is late or fails inspection, offer a verified, suitable higher-tier bike at the original price with the customer's agreement, if available. Otherwise, offer another time or branch.
Would revisit	Revisit if owner policy, observed late-return patterns, staff workflows, or booking-race tests change the feasibility or risk of the trade.

AI Use Record

Team member	Minnette Johnson — UC-01 (preparation draft, Nov. 3, 2026)
Tool	ChatGPT (GPT-6)
Technique	Week 3: "Ask for Every Reading" – enumerate intended and alternative interpretations, classify and tighten steps.

Prompt	For each UC-01 draft step, enumerate all intended, wrong-but-plausible and implausible readings; recommend tighter wording, citing v1.0 and Worrell A1–A15: 1 Open Fleet at the store; 2 Filter by type and size; 3 Show bikes at this store; 4 Check availability; 5 Show available count; 6 Review e-bike battery; 7 Tell customer they can book.
Excerpt of what came back	Ambiguities: logged-in branch vs requested branch; OR vs AND filters; home store vs verified location; physical presence vs rentability; global vs branch count; stale telemetry; booking vs lookup. Step 4 also risks reliance on a visual status badge.
Kept, changed or rejected, and why	Rewrote 7 draft steps (1–7); kept staff type/size search, current location, multi-rule eligibility, local count and clear booking handoff. Rejected unsupported predicted-return guarantees; Worrell A10 requires post-return checks and FR-06 enforces telemetry limits. See Use Cases §5.
Team member	*Peter Olayiwola*
Tool	*Tool and date: OpenAI ChatGPT, Nov. 4, 2026. Exact model version not recorded.*
Technique	*Ask for Every Reading on UC-02 (Reservations).*
Prompt	*Recorded prompt: "For each of these eight UC-02 steps, list intended and alternative readings; classify intended, wrong but plausible, or not plausible, then rewrite ambiguous steps without adding owner-approved policies."* *Source input: Eight draft UC-02 steps covering booking details, availability, selection and rates, saving, deposit, confirmation, inventory and notification, and customer communication. The analysis was checked against v1.0 and the owner replied A1–A15.*
Excerpt of what came back	*Recorded output excerpt: Step 2 may imply that a due-back time guarantees readiness; step 7 may allow notification before the database commit.*
Kept, changed or rejected, and why	*I rejected the interpretation that an expected return is enough to confirm a reservation. I kept the requested-period eligibility check because a due-back time doesn't prove the bike will be returned on time or pass inspection. I also rejected sending a confirmation before the booking is saved, because the customer could receive a confirmation for a reservation the system never recorded.*
Team member	*Carolann Mora*
Tool	*Claude (model ID claude-opus-5-5), in Claude Cowork*
Technique	*Week 3: "Ask for Every Reading" on UC-03 (Nov. 4, 2026): list the readings of each step, classify them, and rewrite steps with wrong-but-plausible readings.*
Prompt	*For each step of this use case, list every way the step could be read. For each reading, say whether a developer could plausibly build it that way. Use case UC-03, Record Return, Inspect, and Release Bicycle:* *1. The Rental Agent identifies the returned bicycle by its fleet ID, and the system retrieves that bicycle's active rental.* *2. The system closes the rental and records the return time, the receiving store and whether the return was late against the due-back time.* *3. The system sets the bicycle's current location to the receiving store and its status to Returned – Awaiting Check, a status that is never counted as rentable.* *4. The Rental Agent cleans the bicycle, performs the quick safety check and records each check item as passed or failed.*

	5. If the bicycle is an e-bike, the system shows its battery level, any blocking fault and the time of its last telemetry reading. *6. The Rental Agent confirms that every check item passed, and the system records who completed the check and when.* *7. The system re-tests rentability: no workshop hold, no e-bike blocker under FR-06, and no conflicting rental or reservation. If none applies, it sets the status to Rentable at the receiving store.* *8. The system makes the new status visible at all four stores, and only then includes the bicycle in UC-01 lookups and UC-02 period checks.*
Excerpt of what came back	*46 readings across 8 steps. Examples:* *"1B. 'Active' means only rentals still within their period, so an overdue rental … is not found and the return fails."* *"6A. One 'all passed' confirmation can be submitted even if an individual item was marked failed."* *"7F. An e-bike below 30% stays in Awaiting Check even though the check passed, so 'checked but charging' can't be told apart from 'not yet checked.'"* *"8D. The bike is excluded from UC-02 period checks until it's Rentable … it contradicts Conflict 1, option 4."* *The tool noted that the most frequent readings concerned when a status takes effect and who did the work.*
Kept, changed or rejected, and why	*Classified 15 readings intended, 27 wrong but plausible, 4 not plausible; rewrote all 8 steps (see Use Cases §5).* *Kept: rental closes at identification; bike counts in physical stock but not as rentable; failed item creates a workshop hold; telemetry is informational at step 5; status takes effect at the central commit.* *Changed: overdue rentals must be found (1B); one-hour overdue rule from Worrell A12 (2D); home store unchanged (3B); confirmation only by the person who did the check, all items passed (6A/6C); new Checked – Not Rentable status (7C/7F); fresh telemetry and re-test when a blocker clears (5C/7G).* *Rejected: "any future reservation blocks Rentable" (7A) and "excluded from UC-02 even as projected" (8D), both against Conflict 1 option 4. Not adopted as fact: who performs cleaning; kept open for the Week 4 forum.*

Named Ownership

Team member	Owned this week
Minnette Johnson	Document integration / deliverable assembly; Sections 1 and 2; UC-01; UC-07/UC-08 brief cases; own AI review and UC-01 architecture row.
Peter Olayiwola	*Use-case diagram (§3), UC-02 and UC-06, UC-02 AI review and architecture assessment, and Conflict 1.*
Carolann Mora	*UC-03 (fully dressed); UC-04, UC-05 and UC-09 brief cases; own AI review (UC-03) and the UC-03 architecture row; Conflict 2; Section 8 change log, audit findings and repository commit.*